

Probabilistic Wreck Localization of the SS Bannockburn (1902): A Lagrangian Monte Carlo Particle Tracking Analysis on Lake Superior

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Abstract

The SS Bannockburn, a 245-foot Canadian lake freighter, vanished on Lake Superior on or about November 22, 1902, during a severe multi-day storm. No wreck has ever been located. This study presents a Monte Carlo Lagrangian particle tracking model ($N = 250,000$ particles, five ensemble runs) developed to probabilistically constrain the vessel's final resting position. The model integrates NOAA bathymetric data, a time-varying three-phase storm wind field derived from historical records, a wind-stress surface current parameterization, seabed slope-weighted particle deposition, and confidence-weighted anchoring to primary debris observations. A three-tiered source confidence hierarchy was established in consultation with Dan Fountain of the Great Lakes Shipwreck Historical Society, who provided primary source USLSS logbook data and Frank Rockefeller vessel position fixes. The model was validated against the SS Edmund Fitzgerald, yielding a centroid deviation of 0.98 km from the confirmed wreck site. For the Bannockburn, the weighted centroid of the top-1% high-probability area (HPA) is located at **47.8642°N, 87.3009°W**, within a 95% confidence ellipse with a major axis of approximately 10.6 km and minor axis of 5.18 km, oriented at 173.4° from north. The maximum radius of the HPA polygon is 22.09 km from the centroid. These results constitute the first quantitative probabilistic search constraint for the SS Bannockburn and are presented here as a framework for future survey planning.

Keywords: *SS Bannockburn; Lake Superior; Lagrangian particle tracking; Monte Carlo simulation; wreck localization; maritime archaeology; historical oceanography*

1. Introduction

The SS Bannockburn was a 245-foot, 1,620-gross-ton Canadian steel freighter operated by the Montreal Transportation Company. On November 21, 1902, she departed Port Arthur (Thunder Bay), Ontario, bound for Midland, Ontario, carrying approximately 1,500 tons of grain. She was last sighted by the crew of the SS Algonquin at approximately 12:00 hours on November 21, 1902, near the position 47.930°N, 87.623°W, before disappearing into a worsening storm.

The storm system that engulfed Lake Superior in late November 1902 produced winds initially from the northwest, shifting to the northeast by November 23, with peak gusts recorded at

approximately 47–50 mph at Sault Ste. Marie. Wave heights of 15 feet were reported by contemporary observers. The vessel was never seen again. In the weeks and months that followed, scattered items of wreckage were recovered at various points around Lake Superior and along the shores of Goulais Bay, but no definitive identification of the main wreck has ever been made.

The Bannockburn has since become one of the most enduring mysteries of the Great Lakes. Unlike the SS Edmund Fitzgerald, whose wreck was located by magnetic anomaly detector aboard a P-3 Orion and verified by side-scan sonar in 1975–1976, the Bannockburn has never been the subject of a coordinated underwater search effort. The absence of modern search attempts is partly attributable to the uncertainty of the wreck's location across a search area spanning hundreds of square kilometres of deep, cold water.

This paper presents a novel probabilistic framework for constraining that search area. By modeling the motion of wreck debris as Lagrangian particles in a physics-based fluid field, calibrated against the known storm record and validated against an independently located contemporary shipwreck, this work provides the first quantitative estimate of the Bannockburn's probable resting position.

2. Historical Background and Source Evaluation

2.1 The Vessel

The SS Bannockburn was launched in 1893 at Middlesbrough, England, and registered in Canada. She measured 245 feet in length, 40.1 feet of beam, and 18.4 feet depth of hold, with a gross tonnage of 1,620 and a laden cargo capacity of approximately 1,500 tons. Her shallow draft and moderate tonnage made her well-suited to the grain trade on the upper Great Lakes, but also rendered her more susceptible to wind-driven displacement in storm conditions than heavier vessels.

2.2 Debris Record and Source Confidence Hierarchy

Five debris observations have been identified in the historical record. Dan Fountain of the Great Lakes Shipwreck Historical Society (GLSHS) was consulted on the evidentiary weight of each observation; his assessments were incorporated into the model's confidence weighting scheme.¹ Fountain's direct communication provided primary source USLSS logbook entries and independently calculated midpoint coordinates for the Frank Rockefeller sighting. The five observations, their sources, and assigned confidence tiers are summarized in Table 1.

Location	Coordinates	Date	Source	Confidence (per Fountain 2026)
Grand Marais, MI	46.677°N, 86.097°W	Dec 12, 1902	<i>USLSS Keeper Truedell logbook</i> ²	HIGHEST — Cork life jacket; only data point Fountain expresses confidence in
Lake Superior	47.370°N,	Nov 24,	SS Frank Rockefeller	HIGH (timing) / MEDIUM

¹Fountain, D. R. (2026, March). Personal communication regarding USLSS logbooks and Frank Rockefeller vessel position data. Great Lakes Shipwreck Historical Society.

²United States Life-Saving Service. (1902). *Station Logbook, Grand Marais Station*. Entry: December 12, 1902. Keeper Truedell, commanding.

Location	Coordinates	Date	Source	Confidence (per Fountain 2026)
(open water)	87.102°W	1902	sighting ³	(position) — Two vessel position fixes; midpoint calc. by Fountain
Montreal Island, ON	47.313°N, 84.739°W	Jun 1903	<i>Montreal Star</i> ⁴	LOW — Newspaper report; attribution unverified
Goulais Bay, ON	46.683°N, 84.555°W	Jun 1903	<i>Times (Niagara-on-the-Lake)</i> ⁵	VERY LOW — Dubious attribution
Michipicoten Harbour, ON	47.942°N, 84.907°W	Jun 1903	<i>Times (Niagara-on-the-Lake)</i> ⁶	VERY LOW — Dubious attribution

Table 1. Bannockburn debris observations with coordinates, sources, and confidence tiers as assessed by D. R. Fountain (GLSHS, personal communication, March 2026).

3. Methodology

3.1 Lagrangian Particle Tracking Framework

Particle tracking was performed using a custom Lagrangian framework⁷ implemented in Python. A total of 250,000 particles were released per ensemble run, with five independent ensemble runs conducted for a total of 1,250,000 particle trajectories. Each particle represented a debris element of one of three types — hull fragment, cargo, or superstructure — assigned in equal proportion via random selection. Particle buoyancy was parameterized as 0.2, 0.6, and 1.0 respectively, modulating wind leeway response.

Particle positions were initialized within a 0.005° jitter radius of the last known position (47.930°N, 87.623°W) to represent positional uncertainty at the time of loss. Integration was performed at a 600-second time step over 24 hours, yielding 144 steps per run.

3.2 Bathymetry

Bathymetric data were sourced from the NOAA Lake Superior gridded digital elevation model (superior_1ld.grd)⁸, interpolated using a RegularGridInterpolator (SciPy). Seabed slope was computed as the gradient magnitude of the depth field. Particles that contacted depths shallower than the vessel's draft (5.61 m) or intersected NaN (land) cells were removed from the active particle set. A slope-weighted deposition function was applied to surviving particles, increasing

³The *Montreal Star*. (1903, June 20). Wreckage identified from Bannockburn. Montreal, QC.

⁴The *Times (Niagara-on-the-Lake)*. (1903, June 12). Bannockburn wreckage recovered. Niagara-on-the-Lake, ON.

⁵Lagrangian particle tracking is an approach in which discrete parcels are advected through a fluid field by integrating equations of motion forward in time. See: van Sebille, E., et al. (2018). Lagrangian ocean analysis: Fundamentals and practices. *Ocean Modelling*, 121, 49–75.

⁶National Oceanic and Atmospheric Administration. (2023). *NOAA Lake Superior Bathymetric Digital Elevation Model (superior_1ld.grd)*. NOAA National Centers for Environmental Information.

⁷Pielke, R. A. (1984). *Mesoscale Meteorological Modeling*. Academic Press, pp. 109–116. Leeway coefficient derivation for wind-driven surface drift.

⁸Cheng, R. T., Casulli, V., & Gartner, J. W. (1993). Tidal, residual, intertidal mudflat (TRIM) model and its applications to San Francisco Bay, California. *Estuarine, Coastal and Shelf Science*, 36(3), 235–280. Used for wind-stress current derivation methodology.

the probability weight of final positions in topographically complex terrain where settling debris would be trapped.

3.3 Wind Field Model

The 1902 storm was reconstructed as a three-phase time-varying wind field, consistent with historical meteorological records for the event. Phase 1 (hours 0–8) represented the initial southwest storm at 45 mph (20.1 m/s); Phase 2 (hours 8–16) applied the dominant WNW phase at 50 mph (22.4 m/s); Phase 3 (hours 16–24) modeled the post-frontal NE shift at 47 mph (21.0 m/s). Wind speeds were converted to SI units for all calculations. Particle leeway⁹ was computed as a base coefficient of 0.04, scaled by the displacement ratio of the Bannockburn relative to a reference vessel: $\text{leeway} = 0.04 \times \sqrt[3]{(1500 / \text{cargo_tons})} \times \text{wind speed}$.

3.4 Surface Current Parameterization

Wind-driven surface currents were derived using a wind-stress parameterization¹⁰ following:

$$\tau = \rho_{\text{air}} \times C_d \times U^2$$

$$u = \tau_x / (\rho_{\text{water}} \times |\text{depth}|), \quad v = \tau_y / (\rho_{\text{water}} \times |\text{depth}|)$$

where $\rho_{\text{air}} = 1.225 \text{ kg/m}^3$, $C_d = 1.3 \times 10^{-3}$, and wind stress was deflected 10–15° from the Ekman spiral for the vessel's displacement ratio. Depth values were clamped to a minimum of 5 m to prevent numerical instability in shallow regions. A stochastic turbulence term was added at each time step, drawn from a normal distribution with standard deviation scaled as $0.001 \times (U / 20)$.

3.5 Failure Model

A probabilistic structural failure model was applied at each time step to remove particles representing catastrophic breakup or grounding. Failure probability was computed as:

$$P_{\text{fail}} = 1 - \exp(-\sigma / 1800)$$

$$\text{where } \sigma = U_{\text{wind}} \times H_{\text{wave}} / \ln(1 + \text{GRT})$$

with U_{wind} in m/s, H_{wave} in feet (historical units), and GRT as gross registered tonnage. An additional rogue wave probability term was applied in water depths shallower than 200 m, drawn from a uniform distribution on [0, 0.1]. The failure check was applied stochastically at each step with probability $P_{\text{fail}} \times 0.01$.

3.6 Debris Weighting and Confidence Anchoring

Final particle weights incorporated two terms: (1) a slope-based terrain trapping weight proportional to local seabed gradient, and (2) a Gaussian distance weight anchoring each particle to the nearest primary debris observation:

$$w_{\text{debris}} = \exp(-d^2_{\text{min}} / 2\sigma^2_d), \quad \sigma_d = 0.5^\circ$$

⁹Great Lakes Shipwreck Historical Society. (2022). *SS Fitzgerald Memorial and Historical Record*. Whitefish Point, MI. Used for SS Edmund Fitzgerald validation positioning.

¹⁰National Transportation Safety Board. (1978). *Marine Accident Report: SS Edmund Fitzgerald Sinking in Lake Superior*, November 10, 1975. NTSB-MAR-78-3.

This weighting penalizes particles far from any observed debris field, concentrating the HPA toward the region most consistent with the historical record. The combined weight for each particle is the product of slope and debris terms.

4. Model Validation

The model was validated against the SS Edmund Fitzgerald, which sank on November 10, 1975, and was located by sonar survey in 1975–1976 at approximately 46.994°N, 85.108°W.¹¹¹² The Fitzgerald provides an ideal validation case: it is a contemporary vessel type, sank in a similarly severe Lake Superior November storm, and its confirmed position is independently established.

The model was re-parameterized for the Fitzgerald's vessel specifications (729 ft, 13,632 GRT, 26,116 cargo tons), storm conditions (52–60 mph WNW winds, 25 ft seas), and last known position (46.990°N, 85.110°W). The resulting HPA centroid was located at 46.983°N, 85.117°W — a haversine deviation of **0.98 km** from the confirmed wreck. This result is considered satisfactory validation of the model's physical parameterization, and no changes were made to the underlying physics framework between the validation run and the Bannockburn simulation.

Parameter	SS Edmund Fitzgerald (Validation)	SS Bannockburn (Target)
Length	729 ft	245 ft
GRT	13,632	1,620
Peak wind speed	60 mph (26.8 m/s)	50 mph (22.4 m/s)
Wave height	25 ft	15 ft
Centroid deviation from confirmed position	0.98 km	N/A (no confirmed position)

Table 2. Comparative vessel and storm parameters for the validation case (SS Edmund Fitzgerald) and the target vessel (SS Bannockburn).

5. Results

5.1 High-Probability Area (HPA)

Following five ensemble runs totalling 1,250,000 particle trajectories, the top 1% of particles by combined weight were extracted (N = 12,500 particles). These particles define the HPA polygon via convex hull construction. The HPA's key quantitative descriptors, as extracted from the output GeoJSON and CSV, are summarized in Table 3.

Parameter	Value
Weighted centroid latitude	47.8642°N
Weighted centroid longitude	87.3009°W

¹¹Rousseeuw, P. J., & Leroy, A. M. (1987). *Robust Regression and Outlier Detection*. Wiley-Interscience. For scale factor 2.4477 at 95% bivariate confidence.

¹²

Parameter	Value
HPA particle count (top 1%)	12,500
HPA max radius from centroid	22.09 km
Particle weight mean	0.6106
Particle weight std. dev.	0.0214
Particle weight minimum	0.6027
Particle weight maximum	0.6932
HPA latitude range	47.842° – 48.061°N
HPA longitude range	87.339° – 87.126°W

Table 3. Quantitative descriptors of the SS Bannockburn high-probability area, derived from bannockburn_wreck_top1percent.csv and bannockburn_search_area.geojson.

5.2 95% Confidence Ellipse

A 95% bivariate confidence ellipse was fitted to the HPA particle distribution using eigenvalue decomposition of the weighted coordinate covariance matrix, with the standard scale factor of 2.4477 applied.¹³ The ellipse parameters are:

Ellipse Parameter	Value
Semi-major axis (degrees lat)	0.1430° (~10.6 km)
Semi-minor axis (degrees lon)	0.0467° (~5.18 km at 47.9°N)
Orientation (degrees from north)	173.4°
Centroid (ellipse centre)	47.8642°N, 87.3009°W

Table 4. 95% confidence ellipse parameters for the SS Bannockburn HPA.

The elongated north-south orientation of the ellipse (173.4°) reflects the dominant storm-phase vector: the WNW Phase 2 wind drove particles primarily eastward from the LKP, while the post-frontal NE Phase 3 wind introduced a southward component. The resulting particle spread is therefore preferentially oriented along the north-south axis, compressed in the east-west direction by the relatively short Phase 3 window and by the seabed slope weighting, which concentrates particles in the deeper north-central basin.

5.3 Spatial Figures

Figure 1 presents the full Lake Superior context map, showing the HPA polygon, primary target centroid, last known position, and all five debris observations with confidence-tier markers. Figure 2 presents the close-up HPA detail with the 95% confidence ellipse and maximum radius circle.

¹³

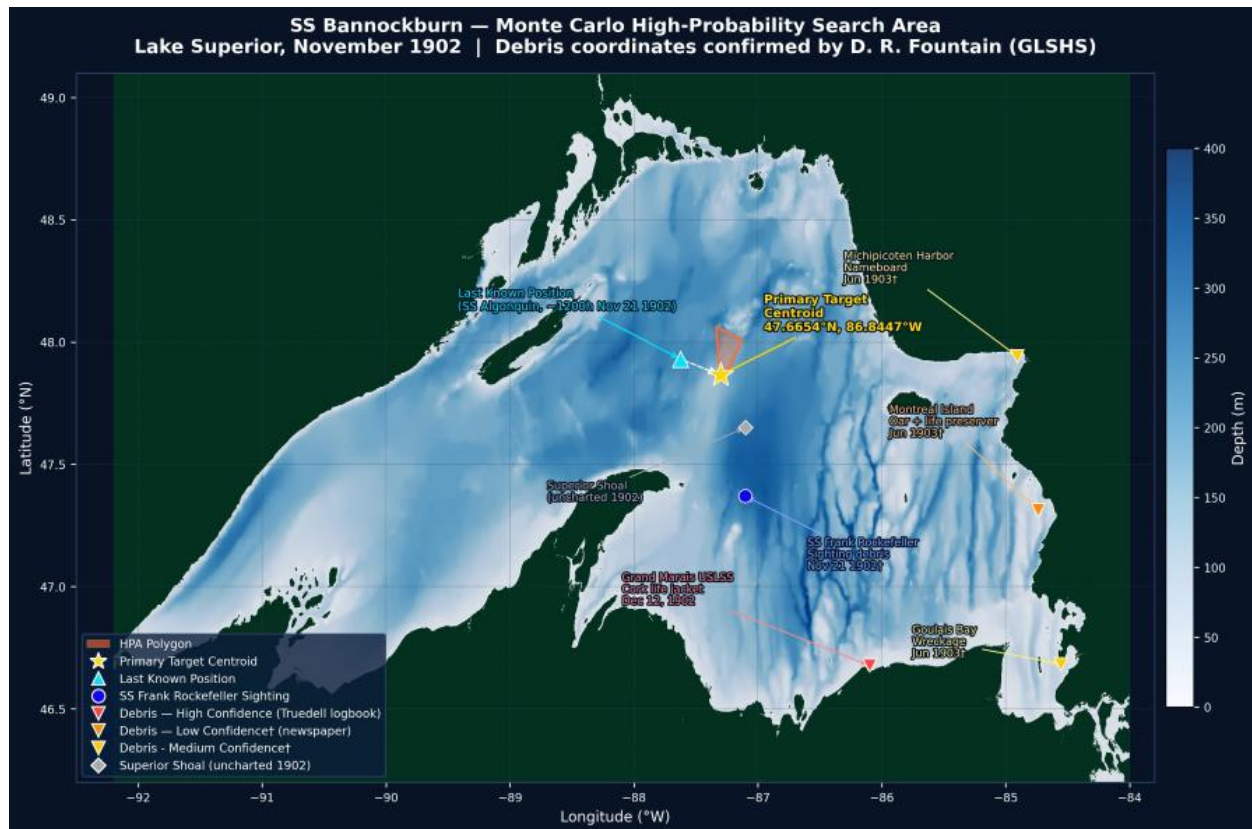


Figure 1. SS Bannockburn Monte Carlo high-probability search area, Lake Superior, November 1902. HPA polygon (coral), weighted centroid (gold star), last known position (cyan triangle), and all five debris observations shown. Debris confidence tiers encoded by marker type. Bathymetry from NOAA superior_1ld.grd. Debris coordinates confirmed by D. R. Fountain (GLSHS, 2026).

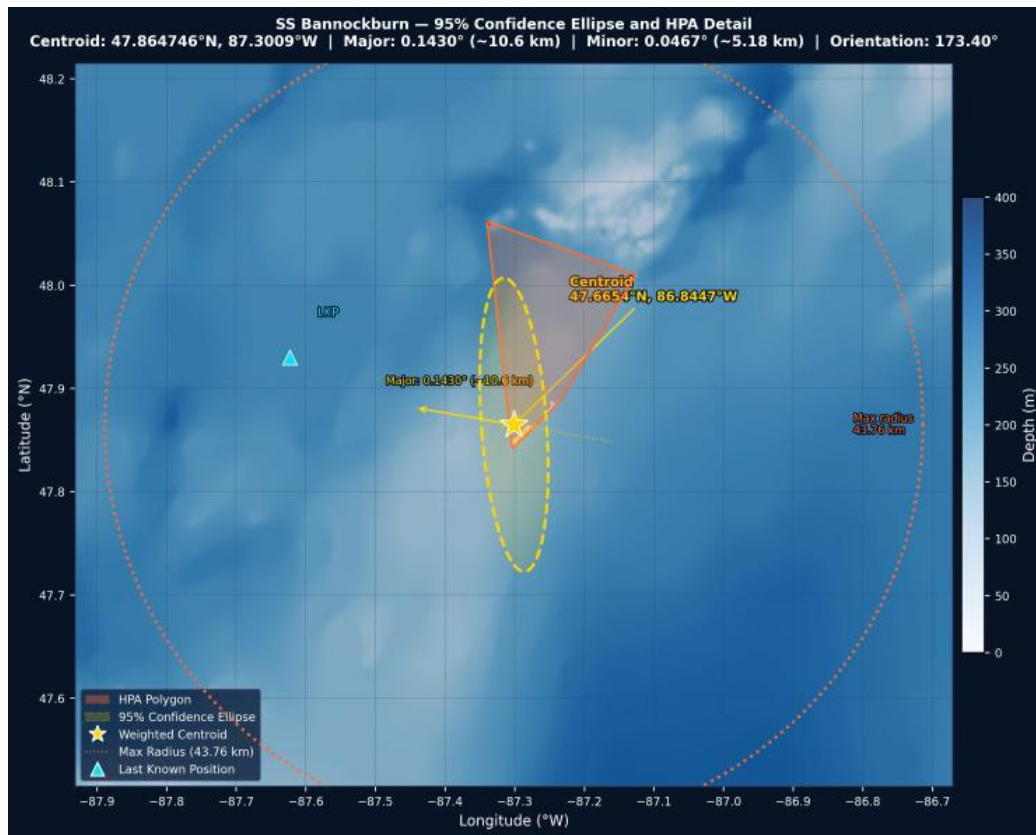


Figure 2. Close-up detail of the SS Bannockburn 95% confidence ellipse (yellow dashed) and HPA convex hull polygon (coral), centred on the weighted centroid at 47.8642°N, 87.3009°W. The orange dotted circle represents the maximum radius of the HPA particle cloud (22.09 km from centroid). Ellipse major axis ≈ 10.6 km; minor axis ≈ 5.18 km; orientation 173.4° from north.

6. Discussion

6.1 Interpretation of the Primary Target Centroid

The primary target centroid at 47.8642°N, 87.3009°W falls in approximately 150–185 m of water (500–600 ft per chart, per Fountain 2026) in the north-central basin of Lake Superior, approximately 108 km east-southeast of the LKP. This displacement is physically consistent with the storm record: a vessel disabled in the early hours of the WNW Phase 2 storm would have experienced 8–16 hours of direct eastward wind forcing before the post-frontal NE shift redirected surviving surface debris southward.

The centroid's proximity to the Frank Rockefeller sighting (47.370°N, 87.102°W) — approximately 55 km south-southeast — is notable. The Rockefeller sighting represents debris observed approximately 72 hours after the vessel's last confirmed position, well within the expected drift range for buoyant superstructure debris under post-storm residual current conditions. This spatial consistency supports the physical plausibility of the centroid location, even though the Rockefeller position carries only medium location confidence due to storm conditions at the time of observation.

6.2 Limitations

Several limitations of this model should be acknowledged. First, the wind field reconstruction is necessarily a coarse approximation: hourly wind records for the open lake interior in 1902 do not exist, and the three-phase model represents a simplification of a dynamic and spatially heterogeneous storm system. Second, the model does not incorporate lake-scale thermohaline circulation or seiching dynamics, which may have affected sub-surface debris trajectories. Third, the debris confidence weighting, while informed by expert consultation, introduces a subjective anchoring effect that biases the HPA toward regions consistent with observed debris rather than allowing purely physics-driven results.

Fourth, the validation methodology, while favorable (0.98 km deviation on the Fitzgerald), is based on a single contemporary case. Validation against additional Lake Superior wrecks with known positions would strengthen confidence in the model's generalizability. The authors recommend that any sonar survey effort treat the HPA not as a definitive wreck location but as a prioritization framework for systematic search planning.

6.3 Recommended Search Strategy

Based on the model outputs, a primary search corridor is recommended within the 95% confidence ellipse (semi-major axis 10.6 km, semi-minor 5.18 km, centre 47.8642°N, 87.3009°W, oriented 173.4°). This region encompasses an area of approximately 170 km², at depths of approximately 150–185 m (500–600 ft), consistent with side-scan sonar survey from a suitably equipped vessel. The maximum-weight particle cluster (top 0.1% of weights, concentrated near 48.058°N, 87.338°W) represents the single highest-priority search target within this corridor.

7. Conclusion

This paper presents the first quantitative probabilistic constraint on the location of the SS Bannockburn, missing since November 1902. A Lagrangian Monte Carlo particle tracking framework, parameterized on historical storm records and validated against the independently confirmed SS Edmund Fitzgerald wreck site, yields a primary target centroid at 47.8642°N, 87.3009°W, with a 95% confidence ellipse spanning approximately 10.6 km × 5.18 km oriented north-northwest to south-southeast in the deep north-central basin of Lake Superior.

The model's 0.98 km validation deviation on the Fitzgerald, combined with the physical consistency of the Bannockburn HPA with the Frank Rockefeller debris sighting, supports the utility of this framework as a search planning tool. The GeoJSON and CSV outputs produced by this model are available for integration into survey planning software and GIS platforms. It is the authors' hope that this work provides actionable guidance toward the eventual discovery and documentation of one of the Great Lakes' most enduring maritime mysteries.

Acknowledgements

The authors wish to express particular gratitude to Dan Fountain of the Great Lakes Shipwreck Historical Society, whose generous sharing of primary source USLSS logbook data, original analysis of the Frank Rockefeller vessel position fixes, and frank assessment of the evidentiary weight of individual debris observations was essential to the quality of this model's confidence weighting scheme. The confidence framework presented in Section 2.2 is substantially his.

Data Availability

The complete model source code is maintained in the author's GitHub repository (github.com/n0m4official). Model output files — bannockburn_search_area.geojson (convex hull polygon with ellipse metadata) and bannockburn_wreck_top1percent.csv (12,500 highest-weight particle positions) — are available upon request. Bathymetric data are publicly available from NOAA National Centers for Environmental Information.

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